

SALIM SOUFI

salim_soufi@icloud.com | linkedin.com/in/salimsoufi | github.com/ssoufi | salimsoufi.com

EDUCATION

University of Alberta

B.Sc. in Computer Science

Edmonton, Alberta

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Product Management, Database Management

EXPERIENCE

Software Engineer Intern

May 2026 – Sep 2026

TD Bank

Toronto, Ontario

- Shipped **React** and **Typescript** features for TD's internal data platform at a **95%** on-time release rate, building **GraphQL** and **REST integrations** against **Collibra** to meet requirements from **20+** data owners, cutting scope-clarification with stakeholders.
- Led **5 developers** rebuilding the Application Inventory platform, refactoring **MUI** front-end flows and **Redux Toolkit** state management across a domain-organized service layer, delivering **60%** faster search for a catalogue of **10,000+** applications.
- Hardened release quality by running weekly **bug bashes** and documenting shared UI patterns in **Storybook**, using **Vite**-based environment configs to safely test against per-mode **Collibra** assets, reducing defects escaping to production.

Software Engineer Intern

Jan 2026 – May 2026

Scotiabank

Toronto, Ontario

- Reduced risk-report generation time by **75%** by engineering **10+** **Python** and **SQL-based automation pipelines** orchestrated via **Apache Airflow DAGs**, delivering **daily risk reports** to a team of **11 developers and analysts**.
- Led development of a **web-based dashboard** consolidating live risk monitors and metadata, integrating with **Azure DevOps** to cut manual reporting by **22+ hours/week** and reduce risk-status turnaround from **days to minutes**.
- Standardized pipeline monitoring by building **automated alerting** into the **Airflow DAGs**, using **Azure DevOps** work items to track failures, shrinking mean time to detect broken pipelines and preventing stale risk data from reaching analysts.

Data Analytics Intern

May 2025 – Dec 2025

TD Bank

Toronto, Ontario

- Ensured **100%** audit-ready compliance by building and maintaining **12+** **Power BI dashboards**, using **Excel** and **Microsoft Office** to clean and visualize data, giving Senior Managers real-time visibility into fund allocation across **12+** projects and departments.
- Analyzed **\$10M in resource allocation** data by consolidating fragmented source spreadsheets, using **Excel** pivot tables and data-validation rules to standardize inputs, cutting recurring data-cleaning time ahead of monthly reporting cycles.

PERSONAL PROJECTS

AI Disaster Response Platform | GitHub

FastAPI, Next.js, Claude API, Twilio, WebSockets

- Built a **disaster alert system** that reaches households through **calls**, **ASL video**, **low-literacy SMS**, or **native-language outreach**, using the **Claude API** to auto-generate alert content per user and **Twilio** to deliver through a **FastAPI** backend.
- Engineered a **real-time WebSocket dispatcher console** in **Next.js** that lets responders monitor alert delivery status across **channels** and **households** live, re-routing failed deliveries to a fallback channel without manual intervention.

LedgerLog | GitHub

LSM-Tree Storage Engine | Python, Log-Structured Storage, WAL, SSTables, Compaction

- Built a **log-structured key-value storage engine** in **Python** for **sequential-write throughput** on audit/event logs: writes land in a **write-ahead log**, then a **concurrent skip-list memtable**, with **crash recovery** replaying the log and halting when torn.
- Recovered read performance with **sparse-indexed SSTables** and **bloom filters** that let lookups skip tables lacking the key, while **size-tiered compaction** merges levels to drop expired/retention-cleared records and bound write and space amplification.

Undergraduate Software Developer | Published Abstract Link

Rehabilitation Robotics Lab, University of Alberta | MATLAB, Kinetisense, OptiTrack

- Published a **peer-reviewed research abstract** validating the **Kinetisense iPad-based motion capture system** through **Sit-to-Stand biomechanical testing**, analyzing **6,700+ motion-capture data points** from **10+ participants** and benchmarking against the **OptiTrack gold standard**.

ACTIVITIES AND LEADERSHIP

Partnership Director

Colorstack UofA

Jan 2026

- Established strategic partnerships with employers through targeted outreach and recruitment coordination, expanding hiring access for **50+** Black and Latinx students.
- Standardized resume collection and distribution workflows, reducing turnaround time for partner requests and improving candidate visibility for underrepresented students.

TECHNICAL SKILLS

Languages: Python, Java, C++, HTML, CSS, JavaScript, TypeScript, Dart, SQL, GraphQL

Libraries & Frameworks: Pandas, NumPy, Django, Flask, React, Express.js, Flutter, Bash

Developer Tools: Jira, Confluence, Apache Airflow, Power BI, AWS, Docker, Node.js, SQLite, MongoDB, REST, Figma, Git

Certifications: Microsoft Azure Fundamentals (AZ-900), Microsoft AI Fundamentals (AI-900), Applied Data Science